

HPE GreenLake for Microsoft Azure Stack HCI

Operations guide

Abstract:

This document supports HPE GreenLake for Microsoft Azure Stack HCI release version 1.2.2023.08.

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Document revision history

Document version	Date	Solution release version
1.0	August 2023	1.2.2023.08



Overview

HPE GreenLake for Microsoft Azure Stack HCI is a cloud-native platform meeting the Microsoft Integrated Systems criteria to provide scalable hyperconverged infrastructure and networking, software, and services in the customer data center. The solution includes an Integrated System Management Plane (ISMP) providing a consolidated management platform for hosting tools necessary for infrastructure and platform management and monitoring.

The following document provide details about proactive support of the solution post release of the solution on the customer premises with details of the updates and patches of each individual component to be installed.

Repository location for Microsoft Azure Stack HCI

The update and patches for all the components in the solution are located at http://elementrepo.us.rdlabs.hpecorp.net/AzureHCI/r2023.08_CPE/.

The patches and updates for respective components are as shown in Table 1.

Table 1. Updates and patches available.

Folder	Purpose
Microsoft Azure Stack HCI update	
Microsoft Windows Server 2022_Patches	
Microsoft Windows Server 2019_Patches	
HPE ProLiant DL 380 Gen10 ROM Flash Firmware Package	
HPE ProLiant DL325 Gen10 PSOC Firmware package	

Refer to the update process for each component as defined in this document along with the update bundle software to update the solution.

Integrated System Management Plane

The Integrated Plane Management (ISMP) server is the control plane of the Microsoft Azure Stack HCI environment. It is a single-node HPE ProLiant DL325 Gen10 Plus v2 server with SUSE Enterprise Linux Edition 15 SP3 as the operating system. KVM Virtualization stack is enabled on the server with the following virtual machines running.

- HPE OneView
- Aruba Fabric Composer
- Windows Admin Center
- Service Now
- OpsRamp
- Metering VM
- Deployment VM
- RDA CAS

Firmware update for HPE ProLiant DL380 Gen10 Plus server

ROM Flash Universal Firmware Package -HPE ProLiant DL360/DL380 Gen10 Plus (U46) servers

1. Log in to the iLO console of the HPE ProLiant DL380 Gen10 Plus server.
2. Select the **Firmware & OS Software** tab.



3. Choose **Select Update Firmware**.
4. Select **Browse** at Local Binary File.
5. Select the firmware file to be flashed.
6. Select **Flash** to begin the update process and wait for the **Firmware flash complete** message to display.
7. Reboot is required after installation of the update.

Firmware update for HPE ProLiant DL325 Gen10 Plus server

PSOC Firmware Flash for Linux(x64) - HPE MR416i-a Gen10 Plus Tri Mode Controller

Obtain the firmware

1. Download the firmware from http://elementrepo.us.rdlabs.hpecorp.net/AzureHCI/r2023.08_CPE/. The Deployment VM or Windows Admin Center VM can be used to connect and download the files.
2. Upload the file to SUSE Linux host using any file upload utility like winscp. Utility. The firmware is to be uploaded to the /opt folder.

Install the firmware

Perform the following steps to install the firmware.

1. Log in to SUSE Linux Host server using user credential as root.
2. Open a terminal window in SUSE Linux Host.
3. Install the firmware rpm using the command: `rpm -ivh firmware-078c22f498-1.0.0.0.1.x86_64.rpm`. This extracts the contents of the rpm to the `usr/lib/x86_64-linux-gnu` location.
4. Navigate to `usr/lib/x86_64-linux-gnu` folder using the command.
`cd /usr/lib/x86_64-linux-gnu`
5. Type "ls" to check the existence of the `firmware-9041739931dr` folder.
6. Navigate to the folder `firmware-078c22f498-1.0.0.0.1.1/`.
7. Run the script.
`./setup.`
8. Once installation is complete, reboot the system.

Updates for virtual machines

Update Microsoft Windows Server 2019 and Microsoft Windows Server 2022

Microsoft Windows Server 2022 and Microsoft Windows Servers 2019 can be updated using one of the following two methods.

- Online update
- Offline update

If the Windows Server has direct connectivity to Public Microsoft Update server, it can download the Microsoft Product updates from it and install the updates. However, in case of no internet connectivity, updates can be downloaded manually from http://elementrepo.us.rdlabs.hpecorp.net/AzureHCI/r2023.08_CPE/ and can be installed as packages to Windows OS.

Refer to the [*HPE GreenLake for Microsoft Azure Stack HCI Firmware and Software Compatibility Matrix - August 2023*](#) to understand the updates that need to be installed.

Online update

To install updates, perform the following the steps.

1. When logged in to a Windows Server operating system, click the **Windows** icon in the task bar and then click **Settings**.



2. After the Windows **Settings** panel is open, locate **Update and Security** and click **Windows Update Section** will open up.
3. Click **check for updates**, the operating system will check of updates and will create a list of all possible updates.
4. Select the updates that are required on the operating system and click **Install**.

If the required updates are not showing up in this method, use the offline method.

Offline update

To install the update manually on a Windows operating system via GUI, follow the steps:

1. Download the update from http://elementrepo.us.rdlabs.hpecorp.net/AzureHCI/r2023.06_CPE/.
2. Place the .msu file on a local folder in the Windows Operating system.
3. Double click the .msu file to execute the update.
4. A **Windows Update Standalone Installer** window will open, where the admin will be prompted to confirm the update installation.

Cluster-Aware updates for Microsoft Azure Stack HCI cluster via Windows Admin Center

Windows Admin Center enables Microsoft Patching, operating system, and hardware updates via simple user interface for integrated systems, Cluster-Aware Updating is an integrated feature that enables updating servers in an Azure Stack HCI cluster with little or no loss in availability during the update process.

Note

Microsoft Azure Stack HCI operating system and SPP updates should be done in accordance to HPE SWFW guide. Security Patches, Security Intelligence Update and other Cumulative Update Packages can be installed as per general guidance.

During the update, Cluster-Aware Updating transparently performs the following tasks:

- Puts each node of the cluster into node maintenance mode.
- Moves the clustered roles off the node.
- Installs the updates and any dependent updates.
- Performs a restart if necessary.
- Brings the node out of maintenance mode.
- Restores the clustered roles on the node.
- Moves to update the next node.

Cluster-Aware Updating is the default updating method used by Windows Admin Center and can also be initiated using PowerShell.

Perform the following steps to install updates.

1. When you connect to a cluster, the Windows Admin Center dashboard alerts you if one or more servers have updates ready to be installed and provides a link to update now. Alternatively, you can select **Updates** from the **Tools** menu at the left.
2. The cluster's update status displays. Click **Check for updates** to get a list of the operating system updates that are available for each server in the cluster. You may need to supply administrator credentials.
3. Select **Next: Install** to proceed to install the operating system updates or click **Skip** to exclude them.



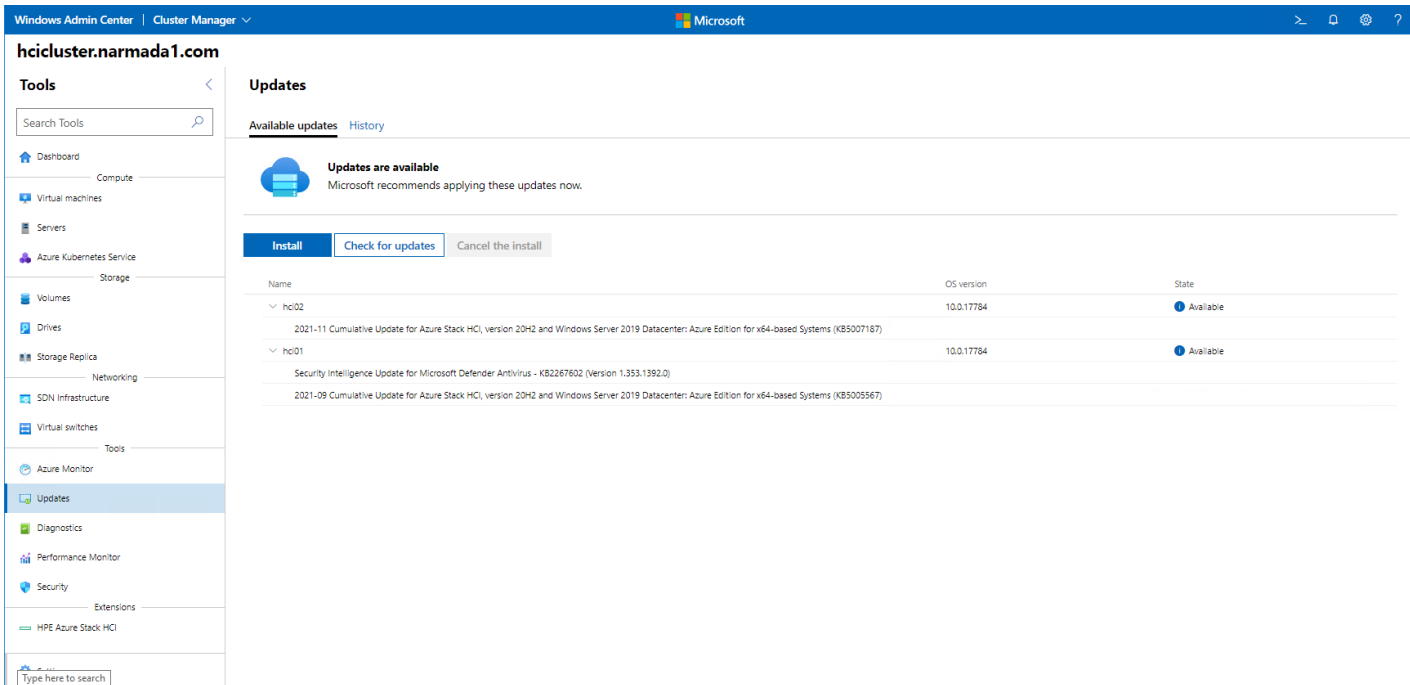


Figure 1. Available software updates

4. When operating system updates are complete, the update status changes to **Succeeded**. Click **Next hardware updates** to proceed to the hardware updates screen.

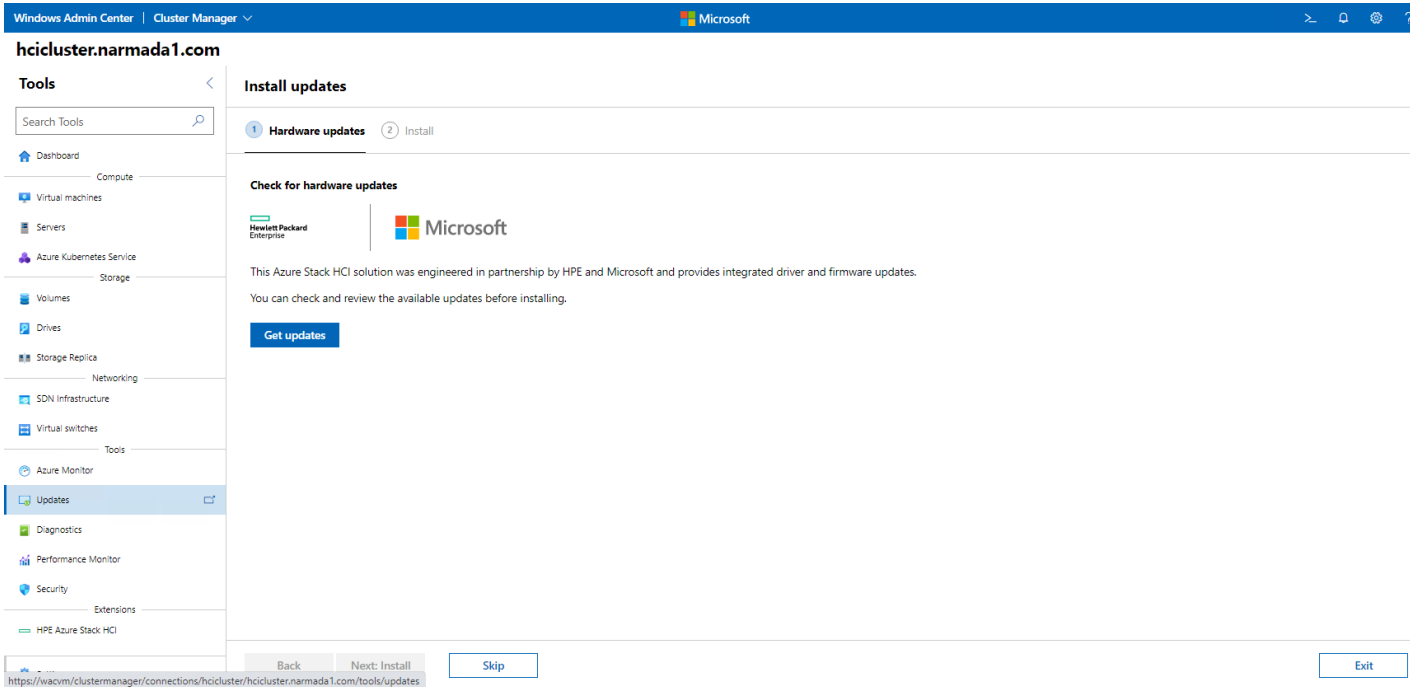


Figure 2. Hardware updates

5. Download the latest SPP, update the path, and click **Continue**.



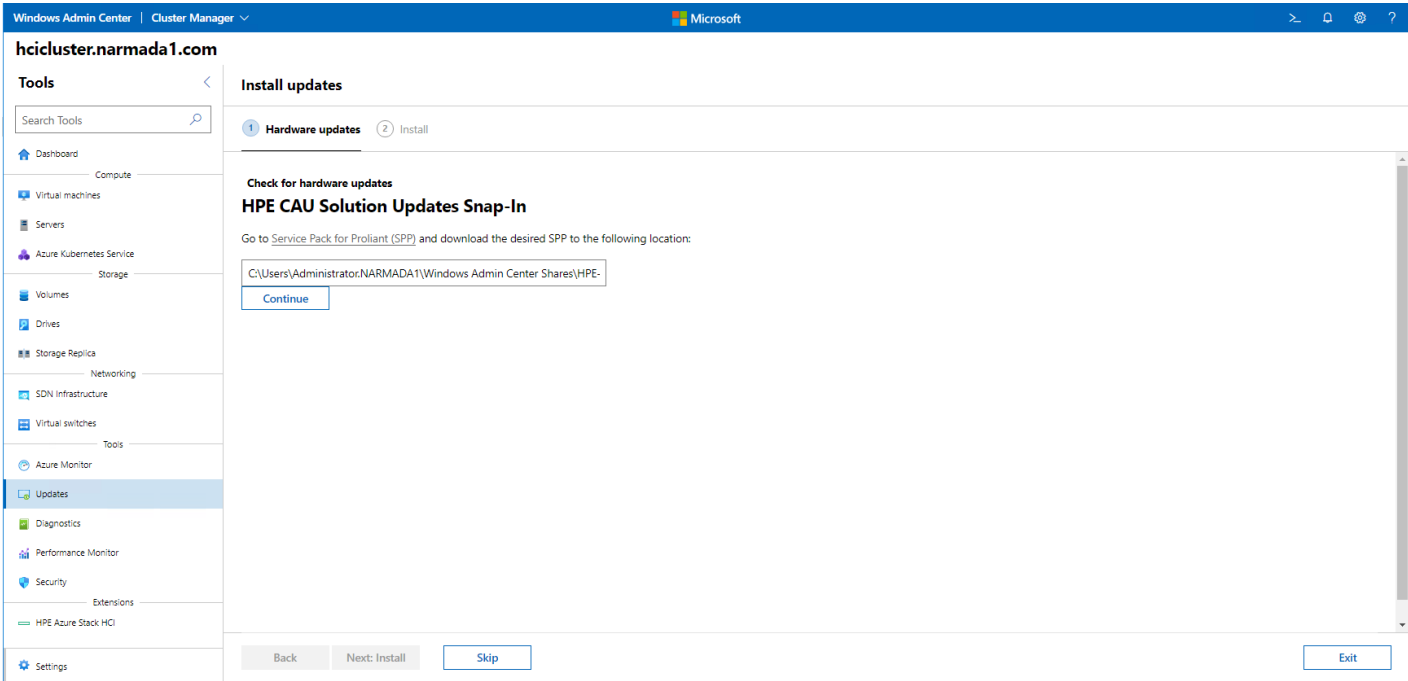


Figure 3. SPP path

6. Select the correct SPP and click **Continue**.

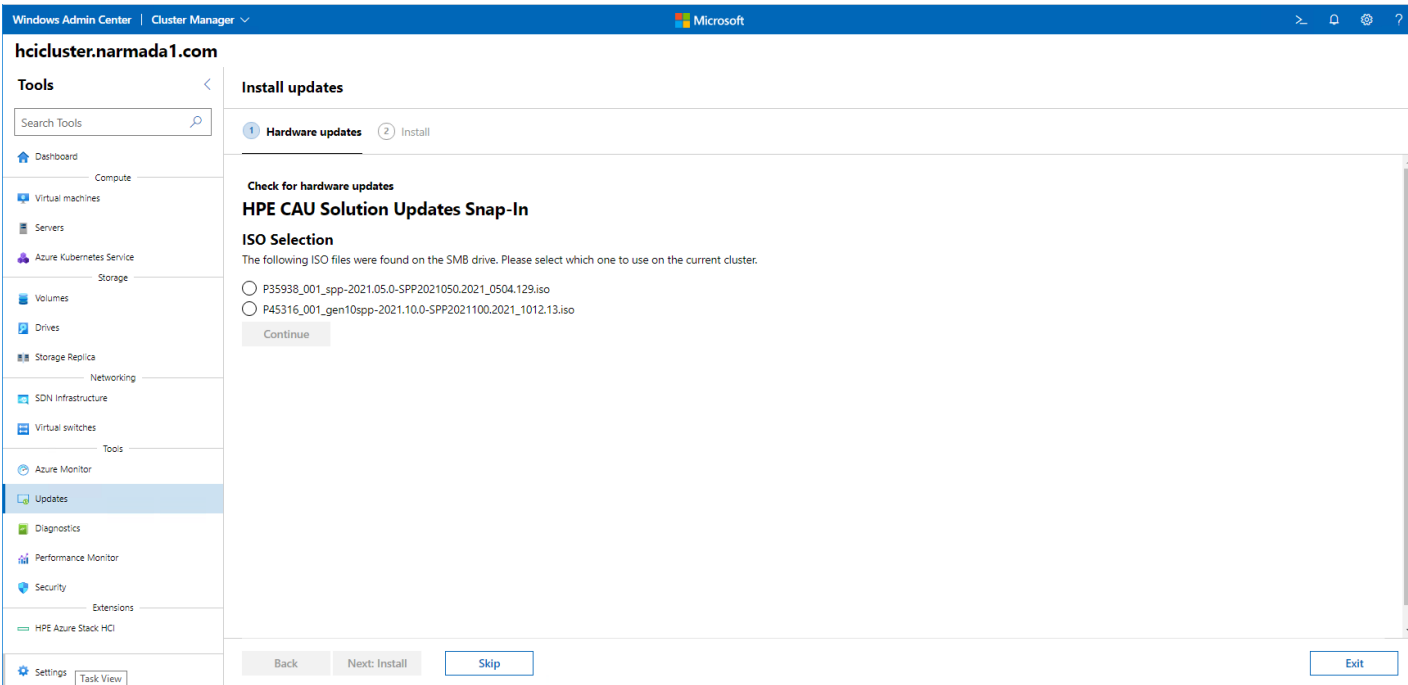


Figure 4. Selecting the SPP ISO from Windows Admin Center UI

Note

You will be able to see the installation progress as in the following figure. Because you are updating the operating system with new features, the updates may take a while to complete. You may be asked to supply your login credentials to Windows Admin Center multiple times.



7. HPE CAU starts copying SPP to nodes. This typically takes around 20 minutes but could be longer if the cluster is having more nodes.

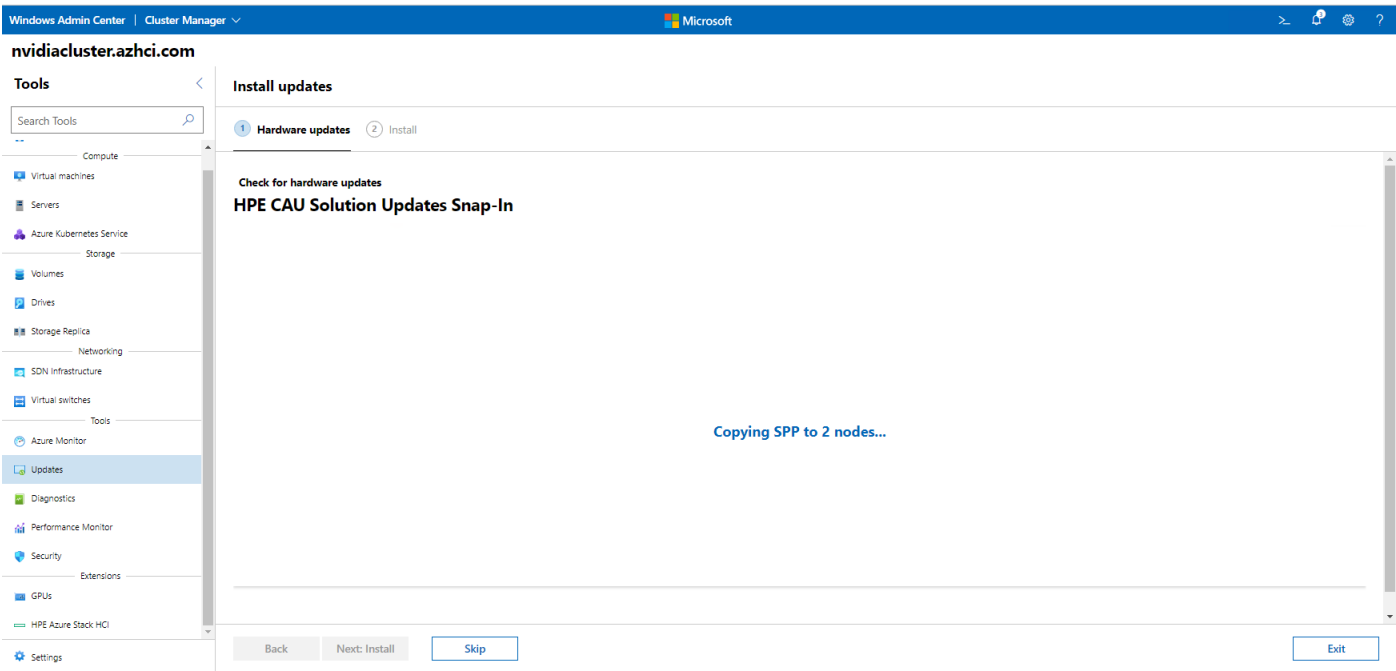


Figure 5. HPE CAU start copying SPP to nodes

8. Available updates for solution nodes are displayed.

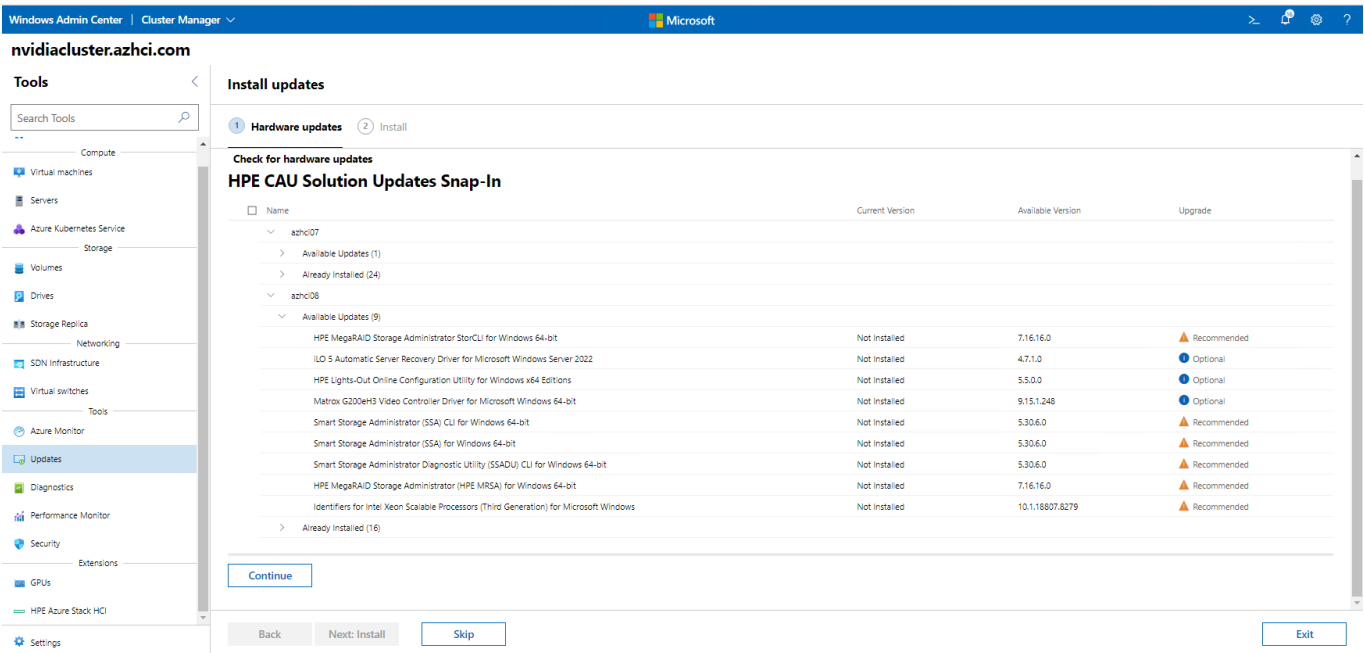


Figure 6. Available hardware updates



9. Start the installation process for updates.

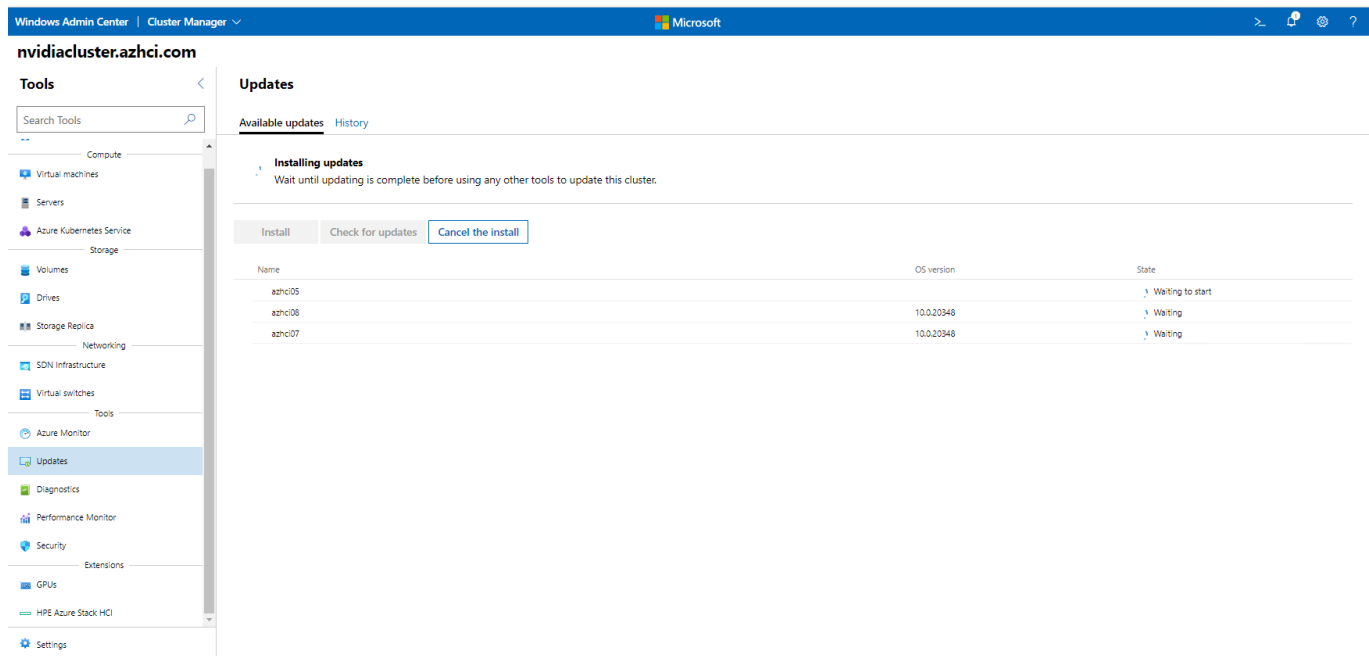


Figure 7. Updates screen showing process starting

10. After the update process is completed, the server restarts without manual intervention.

Updates for PowerShell module

The Microsoft Azure Stack HCI PowerShell module can be updated online. Run the following command with the PowerShell prompt on all the Microsoft Azure Stack HCI cluster nodes to update the PowerShell module.

```
Update-Module -Name Az.StackHCI
```

Solution validation tests

Validation tests must be carried out after a change in hardware component or update to firmware, drivers, or operating systems. The following scenarios should be considered as triggers to execute solution validation tests.

- Windows Server / Microsoft Azure Stack HCI operating system updates
- Windows Admin Center updates
- Driver updates
- Firmware updates
- Switch updates
- Changes in hardware component

We will look at the previous events in brief in the following subsections

Windows Server / Microsoft Azure Stack HCI operating system updates

Updates to Windows components or operating systems initiated from any source (online updates, offline updates, pushed via WAC, manual install, etc.) should be considered as triggers for executing tests to validate solution functionality.



Windows Admin Center updates

Installation of updates to Windows Admin Center or HPE extensions should be considered as triggers for executing tests to validate solution functionality as this impact the management functionality of the solution.

Driver updates

Updates or installation of drivers on Windows operating systems, Microsoft Azure Stack HCI operating system or on other Linux based operating systems should be considered as triggers for executing tests to validate solution functionality.

Firmware updates

Updates or installation of firmware on the HPE ProLiant DL380 Gen10 Plus and HPE ProLiant DL325 Gen10 Plus v2 servers should be considered as triggers for executing tests to validate solution functionality.

Switch updates

Updates to the firmware of Aruba 8325 (production switches) and Aruba 6300M (OOBM switches) should be considered as triggers for executing tests to validate solution functionality.

Changes in hardware component

Changes in hardware components will introduce changes in drivers and firmware, hence they should be considered as triggers for executing tests to validate solution functionality.

The following tests should be run to validate solution functionality:

- Validate iLO health
- Validate server health
- Validate cluster node
- Validate Cluster Health
- Validate Azure Connectivity
- Validate Network Config
- Validate VM Migration
- Validate Secured Core
- Validate connectivity to HPE GreenLake Central

Validate iLO health

To validate the health of iLO for HPE ProLiant DL380 Gen10 Plus and HPE ProLiant DL325 Gen10 Plus v2 servers, perform the following steps.

1. Log in to the iLO webpage using admin credentials.
2. The **Information – iLO Overview** screen appears.



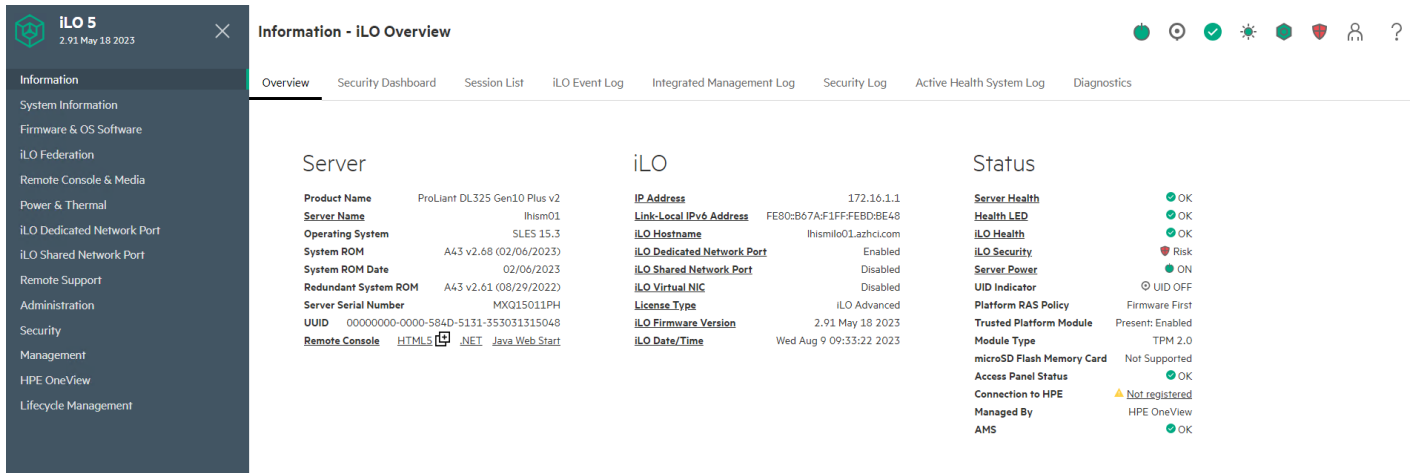


Figure 8. Information – iLO Overview screen

3. Check the status of iLO health indicated on the top right of the webpage by the iLO Health status icon (hexagon). Ensure the iLO health status is green.



4. Click the icon to see the **Information > Diagnostics** screen with iLO Self-Test Results.
5. Ensure the status of all the individual self-tests are green.

Validate server health

To validate the hardware health of the HPE ProLiant DL380 Gen10 Plus and HPE ProLiant DL325 Gen10 Plus v2 servers, perform the following steps.

1. Log in to the iLO webpage using admin credentials.
2. Check the status of the server Health LED icon (circle) and ensure that it is green.



3. Click the icon to navigate to **Information > Integrated Management Log**.
4. If there are any events marked in red on this page, judge its severity/impact and take appropriate action.
5. Check the status of server health indicated on the top right of the webpage by the server Health OK icon. Ensure it is green.



6. Click the checkmark icon to see a drop-down list of server component health status called **Subsystems and Devices**. An overlay with the status of different hardware components such as fans, power, memory, etc., will be displayed. Ensure every component has a green checkmark. If there are any errors or warnings, click the component and you are redirected to a page with more details. Judge the severity/impact and take appropriate action.
7. Click **System Information** in the left pane of the iLO webpage.
8. Browse through all the middle pane options such as processors, memory, storage, etc., and ensure every component has a green checkmark beside it. If any other colors are displayed or if the status shows 'unknown', judge its severity/impact and take appropriate action.



Validate cluster node

To validate the functionality of cluster node or the Microsoft Azure Stack HCI operating system, perform the following steps.

1. Log in to the iLO webpage using admin credentials.
2. On the left bottom of the webpage, click the **remote console** and launch an HTML5 console.
3. Use the **keyboard** icon to send an Ctrl-Alt-Del to the Microsoft Azure Stack HCI operating system.
4. Enter the **admin credentials**, SConfig will load up.
5. Make sure details displayed on the SConfig home screen are correct.
6. Navigate to network and make sure all the IPs, gateways and DNS addresses are in place.
7. Exit back to SConfig home screen and enter powershell.
8. Make sure all the IP address, gateways, and DNS servers are pingable.
9. Make sure nslookup works properly when trying to reach fellow cluster nodes.
10. Now log in to the **Windows Admin Center** via a webpage using admin credentials.
11. On the top right of the Windows Admin Center webpage, click **All Connections** and select **Server Manager** from the dropdown.
12. Select the **server/ cluster node** that needs to be checked, the dashboard for this server will launch.
13. On the left pane of the Windows Admin Center webpage, select **Events**.
14. Check the system, administrative, and Windows events. If any events of concern are found, judge its severity/impact and take appropriate action.

Validate cluster health

Overall cluster health can be validated via the Windows Admin Center by following the steps:

1. Browse to the Windows Admin Center webpage and log in with credentials which have administrative access to the cluster.
2. On the top right corner, expand the drop down and click **Cluster Manager**, a list of available cluster will be displayed.
3. Click the cluster you want to validate, enter the credentials if prompted, the cluster dashboard will load.
4. On the left hand pane in the cluster dashboard click **Servers**.
5. In the middle pane, click **Inventory**.
6. A menu is displayed. Click the three dots as seen in the following image.

7. Click **validate cluster**.
8. Click **Yes** in the validate cluster prompt to start cluster validation.
9. A prompt asking to enable Cred-SSP might show up, click **Yes** to accept and begin cluster validation.
10. After the cluster validation completes, check the validation reports, if any events of concern are found, judge the severity/impact and take appropriate action.

Validate Microsoft Azure connectivity

Microsoft Azure cloud connectivity can be validated via the Windows Admin Center by following the steps:

1. Browse to the Windows Admin Center webpage and log in with credentials which have administrative access to the cluster.
2. On the top right corner, expand the drop down and click **Cluster Manager**, a list of available will be displayed.
3. Click the **cluster** you want to validate, enter the credentials if prompted, the cluster dashboard will load.
4. Azure Connection status should be green and connected as seen in the following image.



Azure connection

Status

✓ Connected ⓘ

Figure 9. Connection Status

5. On the left pane in the cluster dashboard click **Settings**.
6. In the middle pane, scroll down and click **Azure Stack HCI registration**.
7. Scroll down further in the middle pane and click **Sync** as seen in the following image.

Connection

Connection status

✓ Connected

Last connected

7/20/2022, 3:01 AM

This cluster has successfully synced to the cloud within the last day.

Sync

Figure 10. Connection With cloud synced

8. After the action, the cluster will sync again with the Azure cloud and the last connected time stamp will change upon successful connection.

Validate network configuration

Basic network validations on a Microsoft Azure Stack HCI Cluster are done using the Cluster validation tests. Advanced network configurations can be validated via the RDMA Test and DCB validation.

RDMA validation

RDMA validation can be done via a tool called DiskSpd.exe and a PowerShell script named Test-RDMA.ps1. These tools must be downloaded on all the member nodes of the Microsoft Azure Stack HCI cluster and available at the following sources.

- DiskSpd.exe - <https://gallery.technet.microsoft.com/DiskSpd-a-robust-storage-6cd2f223>
- Test-RDMA.ps1 - <https://github.com/Microsoft/SDN/blob/master/Diagnostics/Test-Rdma.ps1>

A directory called **Test** should be created on all the member nodes of the Microsoft Azure Stack HCI cluster to place these two tools. To initiate the RDMA validation the following command is used.

```
C:\TEST\Test-RDMA.PS1 -IfIndex 3 -IsRoCE $true -RemoteIpAddress 192.168.1.5 -PathToDiskspd 'C:\TEST\Diskspd-v2.0.17\amd64fre\'
```

Following are the elements of this command.

- The PowerShell script is downloaded and placed in the directory C:\TEST\; hence to execute it C:\TEST\Test-RDMA.PS1 is used.
- -ifindex is a parameter that requires the index of the network port on which RDMA is supposed to be tested. In the above example the value used is 3, this value will change with the operating system.
- -IsRoCE is a parameter that requires the boolean value of \$true or \$false, this value indicates if RoCE is used. In this solution we use RoCE hence the value \$true is used.
- -RemoteIpAddress is a parameter that requires the IP address of destination port on another node in the Microsoft Azure Stack HCI cluster. The value 192.168.1.5 will change as per the environment.



- PathToDiskspd is a parameter that requires the path to DiskSpd.exe, the value C:\TEST\Diskspd-v2.0.17\amd64fre\ represents the location of DiskSpd.exe.

The output should resemble this:

```
VERBOSE: Diskspd.exe found at C:\TEST\Diskspd-v2.0.17\amd64fre\diskspd.exe
VERBOSE: The adapter NIC1 is a physical adapter
VERBOSE: Underlying adapter is RoCE. Checking if QoS/DCB/PFC is configured on each physical adapter[s]
VERBOSE: QoS/DCB/PFC configuration is correct.
VERBOSE: RDMA configuration is correct.
VERBOSE: Checking if remote IP address, 192.168.1.5, is reachable.
VERBOSE: Remote IP 192.168.1.5 is reachable.
VERBOSE: Disabling RDMA on adapters that are not part of this test. RDMA will be enabled on them later.
VERBOSE: Testing RDMA traffic now for. Traffic will be sent in a parallel job. Job details:
VERBOSE: 0 RDMA bytes written per second
VERBOSE: 0 RDMA bytes sent per second
VERBOSE: 662979201 RDMA bytes written per second
VERBOSE: 37561021 RDMA bytes sent per second
VERBOSE: 1023098948 RDMA bytes written per second
VERBOSE: 8901349 RDMA bytes sent per second
VERBOSE: Enabling RDMA on adapters that are not part of this test. RDMA was disabled on them prior to sending RDMA traffic.
VERBOSE: RDMA traffic test SUCCESSFUL: RDMA traffic was sent to 192.168.1.5
```

Since the member nodes of the cluster in this solution are configured with dual ports for RDMA, this test should be repeated with the second pair of NICs just to ensure the switch configuration is correct.

If this test fails, check to ensure the network switch configuration aligns with the local host configuration.

Validate-DCB

Validate-DCB is a tool provided by Microsoft to validate network settings. HPE GreenLake for Microsoft Azure Stack HCI are configured with remote direct memory access (RDMA) over Converged Ethernet (RoCE) to make the network fabric lossless for storage traffic. This configuration requires Data Center Bridging (DCB) configuration on each server in the cluster and on each network port on the switches through which RDMA traffic passes.

To know more about this tool and to learn how to use it, see <https://docs.microsoft.com/en-us/azure-stack/hci/deploy/validate>.

Validate VM migration

VM migration can be validated via two methods.

Method 1

1. After updates have been completed power on a VM and check the host/node on which it resides.
2. Make sure there are no direct PCIe device assignments on this VM.
3. Right click the **VM** and click **migrate**.
4. Select the **target node** for migration.
5. Click **migrate**.

Method 2

1. Go to **cluster dashboard**



2. Locate inventory and click **Nodes**, all participating nodes of the cluster will be displayed.
3. **Select 1 node** on which there are VMs running with no device assignments.
4. Power down the node from the menu at top of this pane.
5. Click **yes** when asked to confirm VM migration.

Validate secured core

To validate if the secured-core is active on all the Microsoft Azure Stack HCI cluster nodes the following steps must be performed for each node individually:

1. Go to cluster node servers Dashboard.
2. Click **security**.
3. Click **secured core**.
4. Make sure all the six features are set to green.
5. Go to registry and make sure following regkeys are present.
 - a. HKLM\SYSTEM\CurrentControlSet\Control\DeviceGuard\Scenarios\HypervisorEnforcedCodeIntegrity
 - I. Enabled
 - II. Reg_Dword
 - III. 1
 - b. HKLM\SYSTEM\CurrentControlSet\Control\DeviceGuard\Scenarios\SystemGuard
 - I. Enabled
 - II. Reg_Dword
 - III. 1

Resources and additional links

HPE GreenLake for Microsoft Azure Stack HCI Documentation

hpe.com/support/HPE-GreenLake-MS-AzureStack-HCI

Log in to the HPE Support Center as an HPE Employee to view the internal document.

HPE Reference Architectures

<https://www.hpe.com/docs/reference-architecture>

HPE Servers

hpe.com/servers

HPE Storage

hpe.com/storage

HPE Networking

hpe.com/networking

HPE GreenLake Advisory and Professional Services

hpe.com/us/en/services/consulting.html

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Operations guide

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